Bayesian Statistical Inference

Introductory Examples

Edward Milsom

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Slide 1 In this Episode ...

- Recap of probability rules
- Parametric statistical inference: the “orthodox” approach
- Outline of the Bayesian approach
- Examples of Bayesian inference:
  - the mean of the Gaussian distribution
  - the mean of the Bernoulli distribution
- Preview of the lab exercise

Recap: Probability and Bayes' Rule

Slide 2 Recap: Joint & Conditional Probability

Joint probability:

- The probability density (or mass) over the combinations of both values of two variables θ_1 and θ_2 is expressed as: $p\left(\theta_1, \theta_2\right)$

Conditional probability:

- The probability density (or mass) over the values of θ_1 *given* the known value of θ_2 is expressed as: $p\left(\theta_1 | \theta_2\right)$

Variables θ_1 and θ_2 may be either discrete (mass) or continuous (density).

In the context of joint modelling of variables, the quantities $p\left(\theta_1\right)$ and $p\left(\theta_2\right)$ are referred to as the marginal distributions.

Slide 3 Recap: The Rules of Probability

The sum rule (marginalisation):

- If variable θ_2 is an element of some discrete set $\mathcal{S}$ then:

$$P\left(\theta_1\right) = \sum_{\theta_2 \in \mathcal{S}} P\left(\theta_1, \theta_2\right)$$

- If θ_2 is continuous in $\mathcal{R}$ then:

$$p\left(\theta_1\right) = \int_{\mathcal{R}} p\left(\theta_1, \theta_2\right) d\theta_2$$

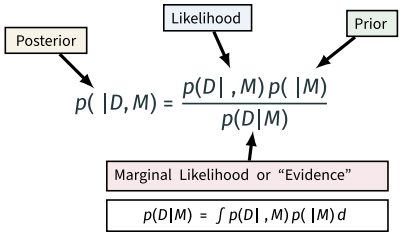
The product rule:

$$p\left(\theta_1, \theta_2\right) = p\left(\theta_1 | \theta_2\right) p\left(\theta_2\right) = p\left(\theta_2 | \theta_1\right) p\left(\theta_1\right)$$

As a consequence of the product rule we get Bayes' rule ...

Slide 4 Recap: The Bayesian Approach

If $\mathcal{D}$ is our observed data, and $\boldsymbol{\theta}$ is the parameter set of a model $\mathcal{M}$ intended to describe it, then the Bayesian approach to "learning" is characterised as:



Parametric Statistical Inference

Slide 5 General ("Orthodox") Parametric Statistical Inference

- A model $\mathcal{M}$ has parameters $\boldsymbol{\theta}$ and we observe a data set $\mathcal{D}$
- We define the likelihood (or observation) model $p\left(\mathcal{D}|\boldsymbol{\theta}, \mathcal{M}\right)$
- Orthodox approach to estimating $\boldsymbol{\theta}$: maximum likelihood

$$\widehat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} \ \log\ p\left(\mathcal{D}|\boldsymbol{\theta}, \mathcal{M}\right)$$

- The "knowledge" gleaned from the data is encapsulated in this single optimised (vector) value, or point estimate, $\widehat{\boldsymbol{\theta}}$
- Predictions for new data $\mathcal{D}_*$ can be made by "plugging in" $\widehat{\boldsymbol{\theta}}$:

$$p\left(\mathcal{D}_*|\mathcal{D},\mathcal{M}\right) = p\left(\mathcal{D}_*|\widehat{\boldsymbol{\theta}},\mathcal{M}\right)$$

Bayesian Statistical Inference

Slide 6 Bayesian Statistical Inference ... on One (Busy) Slide

- A model $\mathcal{M}$ has parameters $\boldsymbol{\theta}$ and we observe data $\mathcal{D}$
- We define the likelihood (or observation) model $p\left(\mathcal{D}|\boldsymbol{\theta},\mathcal{M}\right)$
- The Bayesian approach requires us to define a prior distribution over parameters, $p\left(\boldsymbol{\theta}|\mathcal{M}\right)$, and perhaps a prior over possible models $p\left(\mathcal{M}\right)$
- Inference ("learning") is then the computing of the posterior probability of the parameters given the data via Bayes' rule:

$$p\left(\boldsymbol{\theta}|\mathcal{D},\mathcal{M}\right) = \frac{p\left(\mathcal{D}|\boldsymbol{\theta},\mathcal{M}\right)\, p\left(\boldsymbol{\theta}|\mathcal{M}\right)}{p\left(\mathcal{D}|\mathcal{M}\right)}$$

- The normalising factor $p\left(\mathcal{D}|\mathcal{M}\right)$ is called the marginal likelihood or the evidence (for $\mathcal{M}$)
  - to calculate, we must integrate out $\boldsymbol{\theta}$:
    $p\left(\mathcal{D}|\mathcal{M}\right) = \int p\left(\mathcal{D}|\boldsymbol{\theta},\mathcal{M}\right)\, p\left(\boldsymbol{\theta}|\mathcal{M}\right)\, d\boldsymbol{\theta}$
- We can then make predictions for new data $\mathcal{D}_*$ by averaging over the posterior to give the predictive distribution:

$$p\left(\mathcal{D}_*|\mathcal{D},\mathcal{M}\right) = \int p\left(\mathcal{D}_*|\boldsymbol{\theta},\mathcal{M}\right)\, p\left(\boldsymbol{\theta}|\mathcal{D},\mathcal{M}\right)\, d\boldsymbol{\theta}$$

Slide 7 Bayesian Statistical Inference — Integrating out

Common manipulation of probability distributions in Bayesian inference …

- Perhaps we wish to compute the marginal likelihood $p\left(\mathcal{D}|\mathcal{M}\right)$
- We can write this as:

$$\begin{aligned} p\left(\mathcal{D}|\mathcal{M}\right) &= \int p\left(\mathcal{D},\boldsymbol{\theta}|\mathcal{M}\right)\,d\boldsymbol{\theta} \quad \text{(sum rule)} \\ &= \int p\left(\mathcal{D}|\boldsymbol{\theta},\mathcal{M}\right)\,p\left(\boldsymbol{\theta}|\mathcal{M}\right)\,d\boldsymbol{\theta} \quad \text{(product rule)} \end{aligned}$$

- The latter integrand is the product of the likelihood and the prior, both of which we have specified

Inference: the Gaussian Mean

Slide 8 Example: Mean of the Gaussian Distribution

- The probability density function (p.d.f.) for a single datum:

$$p\left(x|\mu,\sigma^2\right) = \frac{1}{\left(2\pi\sigma^2\right)^{1/2}}\exp\left\{-\frac{\left(x-\mu\right)^2}{2\sigma^2}\right\}$$

- Assume that σ^2 is fixed and known, and μ is to be estimated
- Assume the data, $\mathcal{D}=\{x_1, x_2, ..., x_N\}$, is i.i.d. so:

$$\begin{aligned} p\left(\mathcal{D}|\mu,\sigma^2\right) &= \prod_{n=1}^{N} p\left(x_n|\mu,\sigma^2\right) \\ &= \left(2\pi\sigma^2\right)^{-N/2}\exp\left\{-\sum_{n=1}^{N}\frac{\left(x_n-\mu\right)^2}{2\sigma^2}\right\} \end{aligned}$$

- The maximum likelihood approach to parameter estimation would result in us finding a point estimate for μ, given by $\widehat{\mu} = \bar{x} = \frac{1}{N}\sum_n x_n$

Slide 9 Mean of the Gaussian Distribution — Bayesian Approach

- We need to define a prior distribution over μ — what form should this take?
- Criteria we might consider:
  - Actual prior knowledge (*e.g.* $\mu > 0$)
  - Model structural preference (*e.g.* sparsity)
  - Indifference: a non-informative prior (*e.g.* uniform)
  - Analytic, or computational, convenience (*e.g.* a conjugate prior)
- A conjugate prior is one where the posterior distribution, after applying Bayes' rule for a given likelihood, has the same parametric form as the prior
  - this depends on both the form of the likelihood and the particular parameter being inferred
  - a conjugate prior may not exist!
  - Wikipedia has a useful summary table: https://en.wikipedia.org/wiki/Conjugate_prior

Slide 10 Mean of the Gaussian Distribution — Prior & Inference

For this model, we prefer the conjugate prior for μ, which is also Gaussian and depends on hyper-parameters μ_0 and σ_0^2:

$$p_0\left(\mu|\mu_0,\sigma_0^2\right) = \left(2\pi\sigma_0^2\right)^{-1/2}\exp\left\{-\frac{\left(\mu-\mu_0\right)^2}{2\sigma_0^2}\right\}$$

Then, having observed $\mathcal{D} = \{x_1, x_2, ..., x_N\}$, we apply Bayes' rule to find the posterior $p_N\left(\mu|\mathcal{D}\right)$:

$$p_N\left(\mu|\mathcal{D},\mu_0,\sigma_0^2\right) = \frac{p\left(\mathcal{D}|\mu\right)\,p_0\left(\mu|\mu_0,\sigma_0^2\right)}{p\left(\mathcal{D}|\mu_0,\sigma_0^2\right)}$$

Due to conjugacy, the posterior is also Gaussian.

Useful resource: "Conjugate Bayesian Analysis of the Gaussian Distribution", K. P. Murphy: https://www.cs.ubc.ca/~murphyk/Papers/bayesGauss.pdf

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Slide 11 Mean of the Gaussian Distribution — The Posterior

The posterior is:

$$p_N\left(\mu|\mathcal{D}\right) = \mathcal{N}\left(\mu|\mu_N,\sigma_N^2\right)$$

with

Posterior mean: $\mu_N = \frac{N\sigma_0^2}{N\sigma_0^2+\sigma^2}\bar{x} + \frac{\sigma^2}{N\sigma_0^2+\sigma^2}\mu_0$

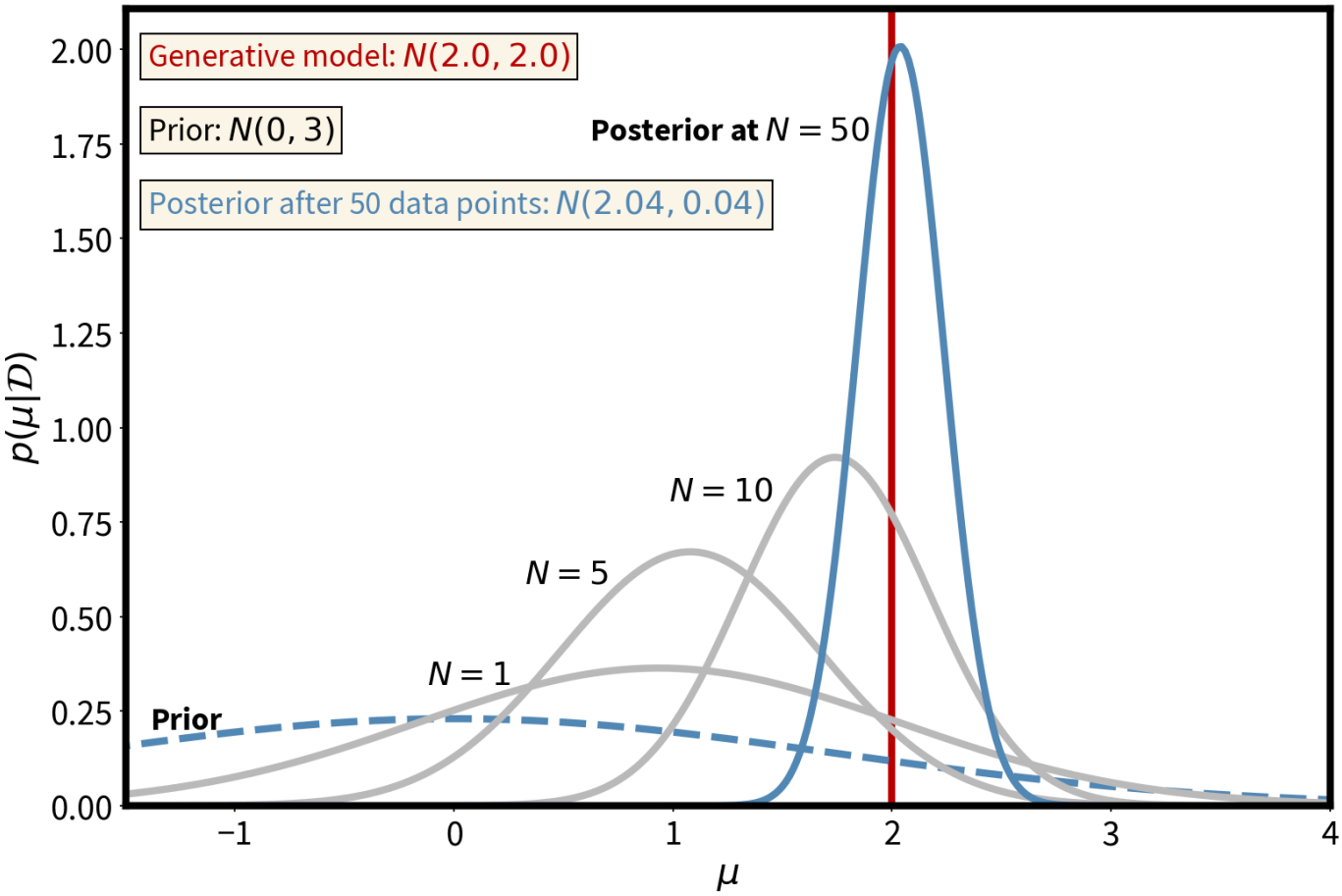
Posterior variance: $\frac{1}{\sigma_N^2} = \frac{N}{\sigma^2} + \frac{1}{\sigma_0^2}$

or precision: $\beta_N = N\beta + \beta_0$, where $\beta_0 = 1/\sigma_0^2$ etc

Note:

- When $N = 0$: $\mu_N = \mu_0$ and $\sigma_N^2 = \sigma_0^2$
- When $N \to \infty$: $\mu_N \to \bar{x}$ and $\sigma_N^2 \to 0$

Slide 12 Illustration: Bayesian Estimation of Gaussian Mean



Slide 13 Mean of the Gaussian — The Predictive Distribution

- Consider we have a new data point $\mathcal{D}_* = x_*$ that we wish to compute the probability of using our Gaussian model
- The Bayesian approach is to compute the predictive distribution:

$$p\left(x_*|\mathcal{D}\right) = \int p\left(x_*|\mu\right) p_N\left(\mu|\mathcal{D}\right) d\mu$$

  - The observation model is: $p\left(x_*|\mu\right) = \mathcal{N}\left(x|\mu, \sigma^2\right)$
  - The posterior is: $p_N\left(\mu|\mathcal{D}\right) = \mathcal{N}\left(\mu|\mu_N, \sigma_N^2\right)$

- It follows that the predictive distribution is also Gaussian:

$$p\left(x_*|\mathcal{D}\right) = \mathcal{N}\left(x_*|\mu_N, \sigma^2 + \sigma_N^2\right)$$

Inference: the Bernoulli Mean

Slide 14 Example: Bernoulli Distribution Probability

- The probability density function (p.d.f.) for a single trial:

$$P\left(x|\theta\right) = \theta^x \left(1-\theta\right)^{1-x}$$

  where $x \in \{0, 1\}$ and θ is probability of "success" ($x = 1$)

- Assume the data, $\mathcal{D} = \{x_1, x_2, ..., x_N\}$, is i.i.d. so:

$$\begin{aligned} P\left(\mathcal{D}|\theta\right) &= \prod_{n=1}^{N} P\left(x_n|\theta\right) \\ &= \theta^{n_1}\left(1-\theta\right)^{n_0} \end{aligned}$$

  where n_0, n_1 denote the number of failures/successes ($x_n = 0, 1$) respectively, with $n_0 + n_1 = N$

- The maximum likelihood estimate is $\widehat{\theta} = \frac{n_1}{n_1 + n_0}$, *i.e.* the observed frequency of "successes"

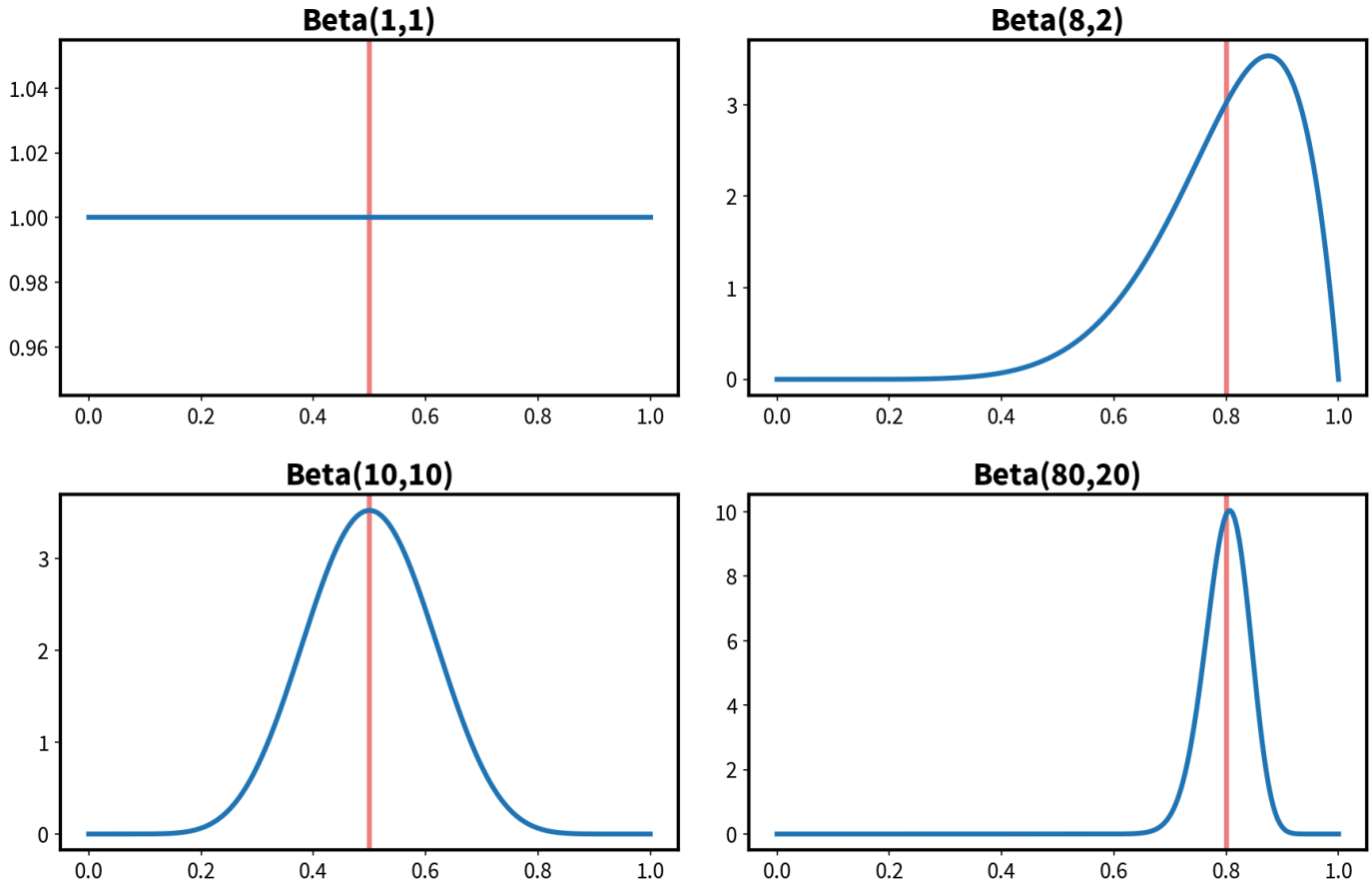
Slide 15 Bernoulli Distribution — the Conjugate Bayesian Prior

The conjugate prior for θ in the Bernoulli is the Beta distribution:

$$p_0\left(\theta|\alpha,\beta\right) = \frac{\Gamma\left(\alpha+\beta\right)}{\Gamma\left(\alpha\right)\Gamma\left(\beta\right)}\theta^{\alpha-1}\left(1-\theta\right)^{\beta-1}$$

- $\Gamma\left(\alpha\right)$ is the Gamma function which appears often in normalisation terms in probability distributions
- The hyper-parameters $\alpha > 0$ and $\beta > 0$ are often referred to as "prior sample sizes" or "pseudo-counts"
- The mean is (intuitively): $E_0[\theta] = \frac{\alpha}{\alpha+\beta}$
- The variance is: $\frac{\alpha\beta}{\left(\alpha+\beta\right)^2\left(\alpha+\beta+1\right)}$

Slide 16 The Beta Distribution



Slide 17 Bernoulli Distribution — Bayesian Inference

Having observed $\mathcal{D} = \{x_1, x_2, ..., x_N\}$, we apply Bayes' rule to find the posterior $p_N\left(\theta|\mathcal{D}\right)$:

$$p_N\left(\theta|\mathcal{D}\right) = \frac{\Gamma\left(\tilde{\alpha}+\tilde{\beta}\right)}{\Gamma\left(\tilde{\alpha}\right)\Gamma\left(\tilde{\beta}\right)}\theta^{\tilde{\alpha}-1}\left(1-\theta\right)^{\tilde{\beta}-1}$$

Due to conjugacy, this is a Beta distribution with:

$$\tilde{\alpha} = \alpha + n_1$$
$$\tilde{\beta} = \beta + n_0$$

Posterior mean:

$$E_N[\theta] = \frac{\tilde{\alpha}}{\tilde{\alpha} + \tilde{\beta}} = \frac{\alpha + n_1}{\alpha + \beta + \left(n_1 + n_0\right)}$$

See: https://en.wikipedia.org/wiki/Conjugate_prior

Slide 18 **Bernoulli Distribution — The Posterior**

Posterior mean:

$$E_N[\theta] = \frac{\alpha + n_1}{\alpha + \beta + \left(n_1 + n_0\right)}$$

So:

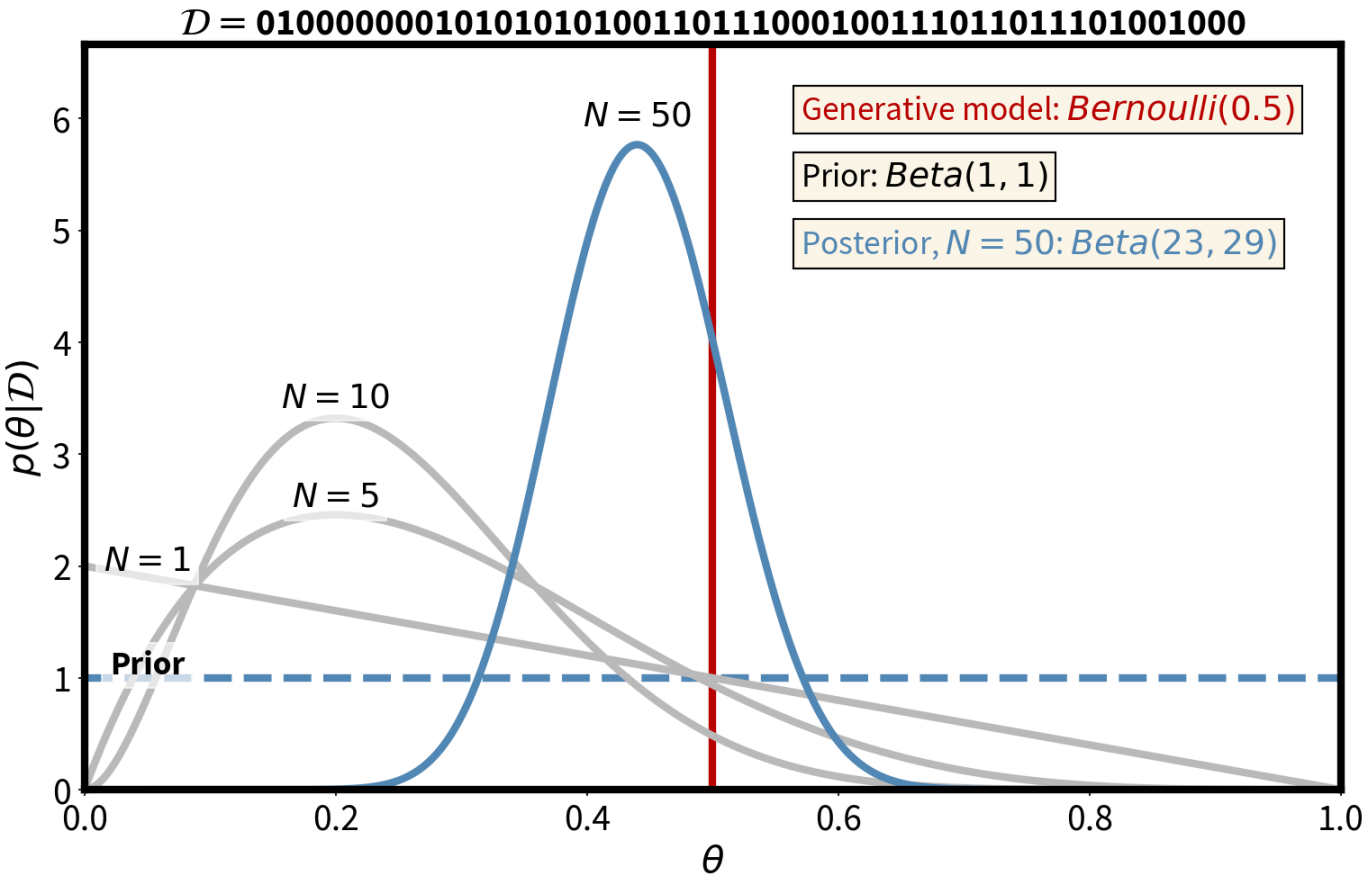
- When n_1 and $n_0 = 0$:

$$E_N[\theta] = \frac{\alpha}{\alpha + \beta}$$

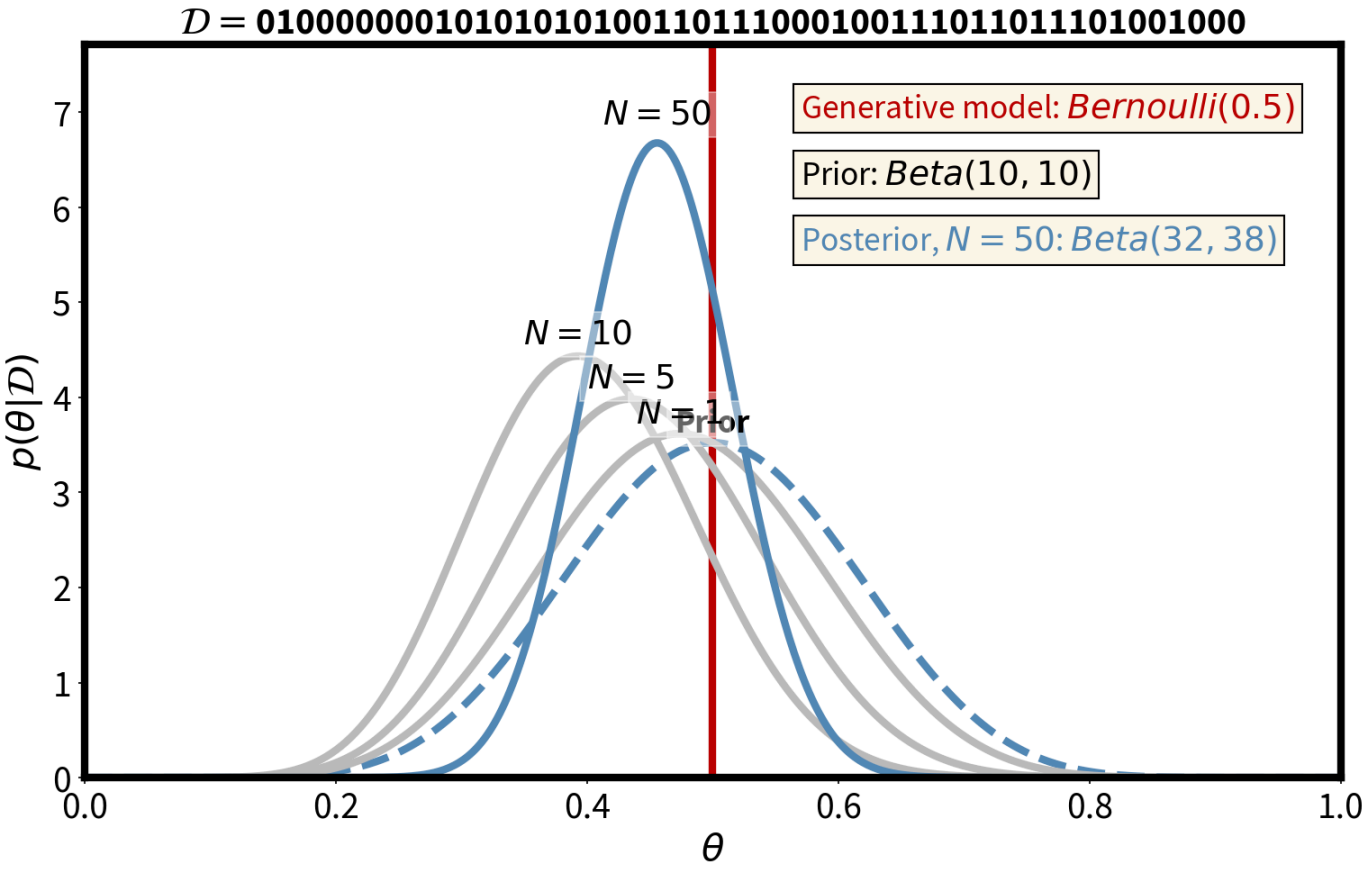
- When $\left(n_1 + n_0\right) \to \infty$:

$$E_N[\theta] \to \frac{n_1}{\left(n_1 + n_0\right)}$$

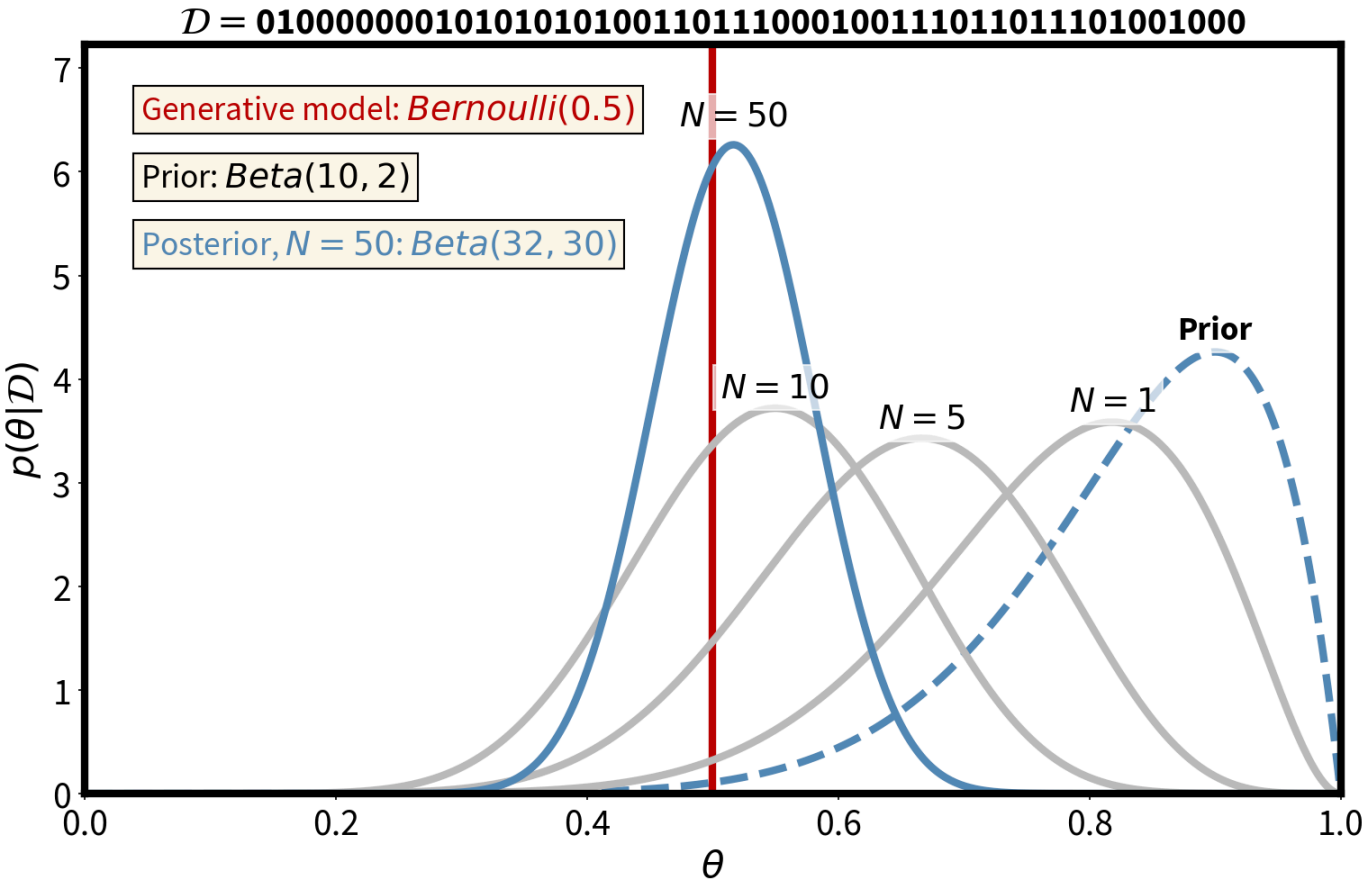
Slide 19 Illustration: Bayesian Estimation of Bernoulli Mean (1)



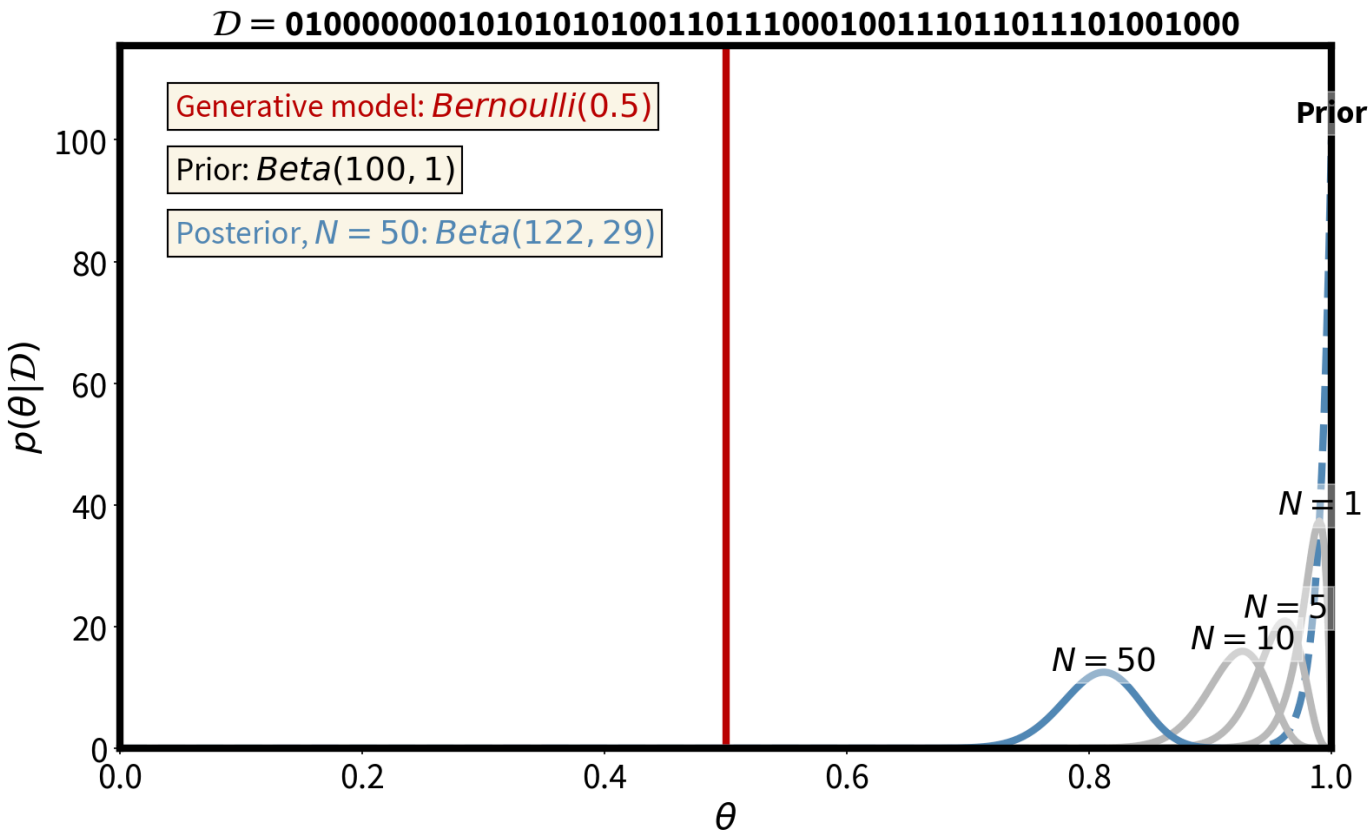
Slide 20 Illustration: Bayesian Estimation of Bernoulli Mean (2)



Slide 21 Illustration: Bayesian Estimation of Bernoulli Mean (3)



Slide 22 Illustration: Bayesian Estimation of Bernoulli Mean (4)



Lab Exercise Preview

Slide 23 Preview: Laboratory Exercise

Tomorrow's lab session will be a warm-up exercise, based around playing a "Multi-Armed Bandit". It will take the form of a Jupyter Notebook, and tasks will include:

- specifying appropriate prior distributions,
- computing corresponding posterior distributions,
- and sampling from them.

Featured observation models will be the Bernoulli and Gamma distributions.

Some familiarity with the *Python* modules `scipy.stats` or `numpy.random` will be useful.